UNIVERSITÉ GRENOBLE ALPES

INTERNSHIP REPORT

Modeling and Control of a PEM Fuel Cell Stack

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Declaration of Authorship

I, Khalil Ellah LAGHA, declare that this thesis titled, "Modeling and Control of a PEM Fuel Cell Stack" and the work presented in it are my own. I confirm that:

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- Where I have consulted the published work of others, this is always clearly attributed.
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"Where there is a will there's a way."

George Herbert

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Abstract

Modeling and Control of a PEM Fuel Cell Stack

by Khalil Ellah LAGHA

This report presents the modeling, calibration, and dynamic characterization of a proton exchange membrane fuel cell (PEMFC) stack carried out at GIPSA-lab. A simplified control-oriented model is developed from electrochemical principles: a Nernst open-circuit voltage combined with semi-empirical activation, ohmic, and concentration losses, coupled to lumped-volume hydrogen, membrane-hydration, and cathode-humidity dynamics. Six parameters per cell are identified from the Optimization Experiment using a two-stage optimization (differential evolution followed by L-BFGS-B), yielding root-mean-square voltage errors below 12 mV for all healthy cells and revealing degraded cells through voltage collapse and degenerate fits. The state-space matrices of the linearized model are written out explicitly — the state and input matrices follow directly from the dynamic equations, while the output matrices require a Jacobian linearization of the nonlinear voltage equation. The system is shown to be structurally controllable for all realistic electro-osmotic drag coefficients, while a voltage-only output observes the hydrogen pressure but not the water states. Step-load experiments (3.3 Ω and 6.8 Ω) characterize the current response of the stack ($\tau_I \approx$ 8–10 s), providing quantitative targets for closed-loop control design. Building on these results, a control architecture based on a DC/DC boost converter regulated by a PI controller is presented, in which the converter output is regulated to behave either as a current source — the approach of Derbeli et al. — or as a voltage source. Results are presented in the body; the open-circuit-voltage derivation and the full per-cell parameter table are gathered in the appendices.

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Chapter 1

Introduction

Fuel cells convert the chemical energy of hydrogen directly into electricity, with water and heat as the only by-products. Among them, the proton exchange membrane fuel cell (PEMFC) is the leading candidate for transport and small stationary applications thanks to its low operating temperature, high power density, and fast start-up [9, 3]. Their efficient and safe operation, however, requires feedback control of the gas supply and load: insufficient reactant flow causes starvation and irreversible degradation, while excessive anode–cathode pressure difference can damage the membrane [16, 6]. Control design, in turn, requires a model that is simple enough for analysis and accurate enough to represent the actual hardware — including the differences between individual cells of a stack [1].

This internship, carried out at GIPSA-lab (Université Grenoble Alpes) under the supervision of Prof. Emmanuel Witrant, addresses this need for a small educational PEMFC stack of ten cells. The objectives are:

1) to develop a simplified, control-oriented dynamic model of a single cell and of the stack, grounded in the electrochemical literature;

2) to identify the model parameters *per cell* from experimental data acquired on the bench;

3) to derive the linearized state-space representation and analyze its structural controllability and observability, as a foundation for controller and observer design;

4) to characterize experimentally the dynamic response of the stack to resistive load steps, providing the timescales needed for control bandwidth selection;

5) to propose a PI-based control architecture for the PEMFC power system through a DC/DC boost converter, following [4].

The report is organized to give a global vision of the work. Chapter 2 summarizes the operating principle of PEMFCs, positions the work in the literature, and explains the practical constraints that shaped the control strategy. Chapter 3 presents the simplified model, the parameter-identification procedure, the linearized state-space analysis, and the experimental protocols. Chapter 4 discusses the calibration and step-response results. Chapter 5 presents the control of the PEMFC power system with a DC/DC boost converter and a PI controller. Chapter 6 concludes and outlines perspectives. The open-circuit-voltage derivation and the full per-cell parameter table are provided in Appendices A and B; the main text is self-contained and readable without them.

Chapter 2

Context & Background

2.1 Operating Principle of a PEM Fuel Cell

A proton exchange membrane fuel cell (PEMFC) converts the chemical energy of hydrogen and oxygen directly into electricity, heat, and water [1, 8, 12]. At the anode, hydrogen splits into protons and electrons. The protons cross the polymer membrane, while the electrons travel through the external circuit and supply the load. At the cathode, oxygen recombines with both to form water (Figure 2.1). The overall reaction is

$$H_2 + \tfrac{1}{2}O_2 \rightarrow H_2O + \text{electricity} + \text{heat}.$$

The membrane conducts protons, blocks electrons, and separates fuel from oxidant. Together with the two electrodes it forms the membrane electrode assembly (MEA), the elementary repeating unit of the stack [1, 5]. Water management is critical to performance: a dry membrane loses proton conductivity, whereas excess liquid water floods the gas channels. Stable operation therefore requires a balance between hydration and drainage [14, 15].

2.2 Voltage Losses and Polarization Behavior

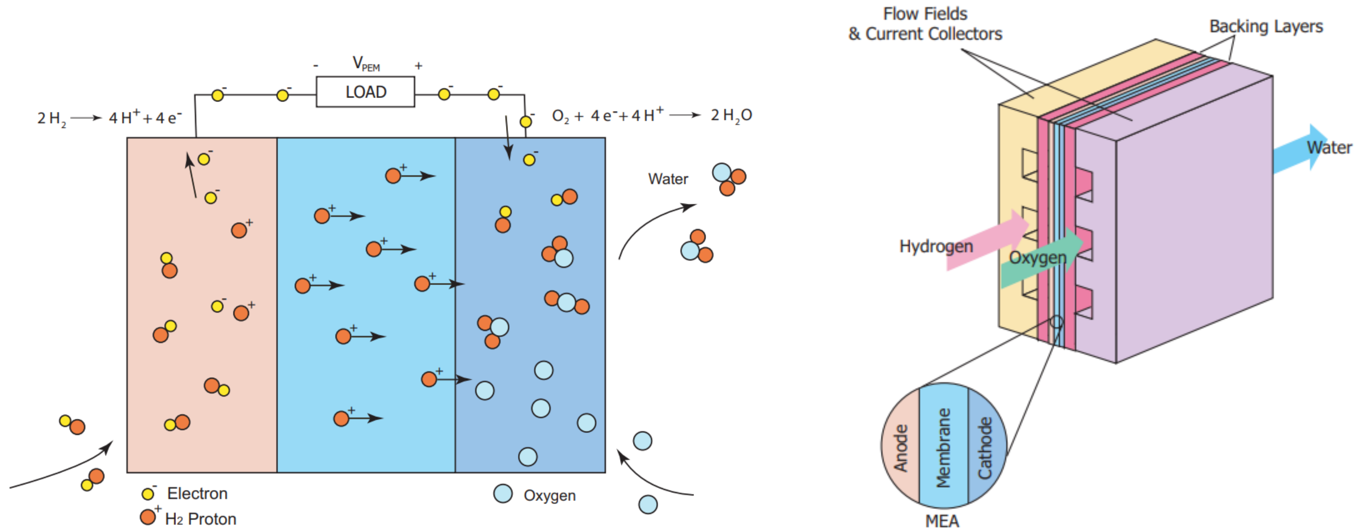
A practical cell delivers less than its reversible (open-circuit) voltage because of three loss mechanisms, each dominant in a different region of the polarization curve (Figure 2.2) [1, 9, 11]:

- **activation loss** — the kinetic penalty of driving the electrode reactions, dominant at low current [9, 14];
- **ohmic loss** — the resistive drop across the membrane and conductors, approximately linear in current [9, 11];
- **concentration loss** — the mass-transport limitation that causes the steep voltage drop at high current [1, 14].

The *polarization curve* is the cell voltage plotted against current density; it is the fingerprint of cell performance and the basis for the model of Chapter 3. Fast transients are further shaped by the charge double-layer at the electrode/electrolyte interface, which behaves as a small capacitor in equivalent-circuit models [9, 1].

2.3 State of the Art

Control-oriented PEMFC modeling traces back to the system-level model of Pukrushpan [9], which couples a polarization-curve voltage law with compressor, manifold-filling, and membrane-hydration dynamics; most later models specialize or simplify it [11, 7, 16]. Simplified electrical models reproduce the static voltage–current characteristic of small stacks with only a handful of parameters [1, 10]. Crucially, per-cell measurements on a small stack reveal voltage deviations of up to 7 % between cells [1], which motivates the per-cell calibration adopted here. Parameter identification by direct-search optimization on measured voltage and power was formalized by Thanapalan et al. [11] and extended with modern metaheuristics [2].



(A) Electrochemical operation: hydrogen oxidation at the anode, proton transport through the membrane, oxygen reduction at the cathode.

(B) Stack structure: individual MEAs stacked in series to add their voltages.

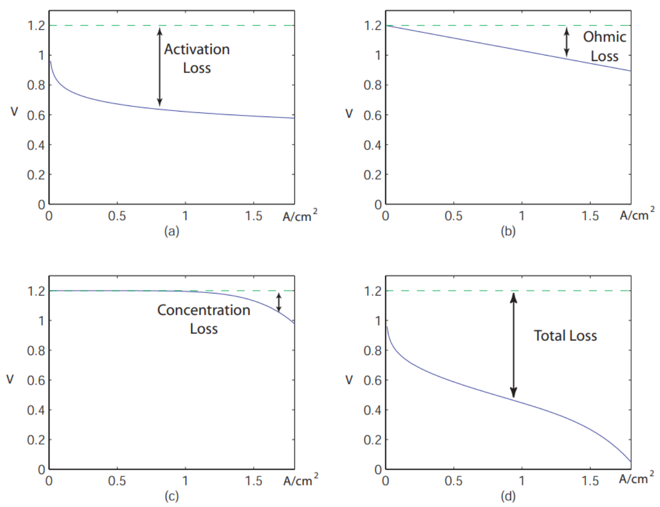
FIGURE 2.1: Operating principle and physical structure of a PEM fuel cell. A single cell delivers roughly one volt, so cells are stacked in series to reach a useful voltage [8, 12, 1].

On the control side, proportional–integral (PI) regulation of flows, pressures, and converter currents is the established baseline [16, 4]. Documented upgrades include linear-quadratic (LQ) multivariable control, time-delay control, and model-predictive schemes [9, 7, 13]. At system level, the response delays of the air and hydrogen supply components remain the main factor limiting dynamic operation [6].

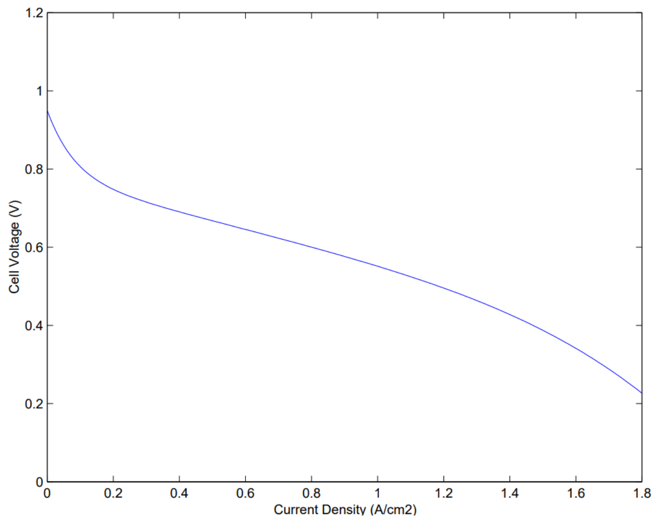
2.3.1 Practical Challenges and Choice of Control Strategy

The control strategy of this work was shaped directly by constraints observed on the laboratory bench. The most natural actuator for a fuel cell is the hydrogen supply: raising the anode pressure raises the Nernst voltage and the available current. In practice, however, the usable hydrogen pressure on the educational stack is narrowly bounded. The supply regulator caps the inlet pressure, and a large anode–cathode pressure difference risks mechanical damage to the thin membrane [16, 6]. The pressure can therefore be varied only within a small range, which yields too little authority to regulate the operating point across the full load. Direct current control through the hydrogen valve was explored and found impractical for the same reason: the actuator saturates before the load demand is met.

This limitation motivated the chosen approach. Rather than acting on the slow, saturation-limited gas supply, the control is moved to the electrical side: the stack feeds a DC/DC boost converter whose duty cycle is set by a fast PI loop [4]. The converter decouples the load from the cell and shapes the delivered power, while the hydrogen supply is left to maintain an adequate pressure rather than to track the load. This separation of roles — electrical control for the fast dynamics, gas supply for the slow operating conditions — underlies the architecture developed in Chapter 5.



(A) Decomposition into activation, ohmic, and concentration losses.



(B) Resulting polarization curve.

FIGURE 2.2: Influence of the three loss mechanisms on PEMFC performance. Activation dominates at low current, ohmic loss in the middle, and concentration loss near the limiting current — together they bend the ideal flat voltage into the characteristic polarization curve.

Chapter 3

Methodology

3.1 Simplified Control-Oriented Model

Following [11, 9], the cell is split into three interacting parts — anode, membrane, and cathode — each lumped into a single control volume. The main assumptions are: hydrogen and air are the only reactants; channel pressure drops are neglected; gases obey the ideal-gas law; temperature is constant or slowly varying; and the oxygen partial pressure is fixed at $P^*_{O_2} = 0.21\, P_{\text{atm}}$. The result is a simplified semi-empirical model: physically motivated mass balances combined with empirical loss laws calibrated from data [2].

3.1.1 Open-Circuit Voltage

The reversible (open-circuit) voltage follows from the Gibbs free energy of the reaction and Faraday's law (full derivation in Appendix A):

$$E = 1.229 - 0.85 \times 10^{-3}(T_{fc} - 298.15) + 4.31 \times 10^{-5}\, T_{fc} \left[\ln p_{H_2} + \tfrac{1}{2} \ln p_{O_2}\right] \ \text{V}, \tag{3.1}$$

with T_{fc} the stack temperature [K] and p_{H_2}, p_{O_2} the reactant partial pressures [atm] [9].

3.1.2 Voltage Losses

The cell voltage is the reversible voltage minus the three losses of Section 2.3:

$$v_{fc} = E - v_{act} - v_{ohm} - v_{con}, \tag{3.2}$$

with the semi-empirical forms [9, 11, 2]

$$v_{act} = \xi_1 + \xi_2 T + \xi_3 T \ln(C_{O_2}) + \xi_4 T \ln(i_{FC}), \tag{3.3}$$

$$v_{ohm} = i_{FC}\Big(\rho_m \frac{t_m}{A_{fc}} + c\Big), \qquad \rho_m = \Big[(b_{11}\lambda_m - b_{12}) \exp\big(b_2(\tfrac{1}{303} - \tfrac{1}{T_{fc}})\big)\Big]^{-1}, \tag{3.4}$$

$$v_{con} = -B \ln\Big(1 - \frac{i_{FC}}{i_{\max}}\Big), \tag{3.5}$$

where i_{FC} is the cell current, C_{O_2} the dissolved-oxygen concentration (Henry's law), t_m and A_{fc} the membrane thickness and active area, λ_m the membrane water content, and $\boldsymbol{\theta} = (\xi_1, \xi_2, \xi_3, \xi_4, B, c)$ the six parameters identified per cell in Section 3.2. The stack voltage is the sum over the n cells in series.

3.1.3 Anode Dynamics

The hydrogen pressure follows a mass balance on the anode volume with Faraday consumption:

$$\dot{P}_{H_2} = \frac{R_{H_2} T}{V_{an}}\Big(W_{H_2,in} - W_{H_2,out} - \frac{M_{H_2} n}{2F}\, i_{FC}\Big), \tag{3.6}$$

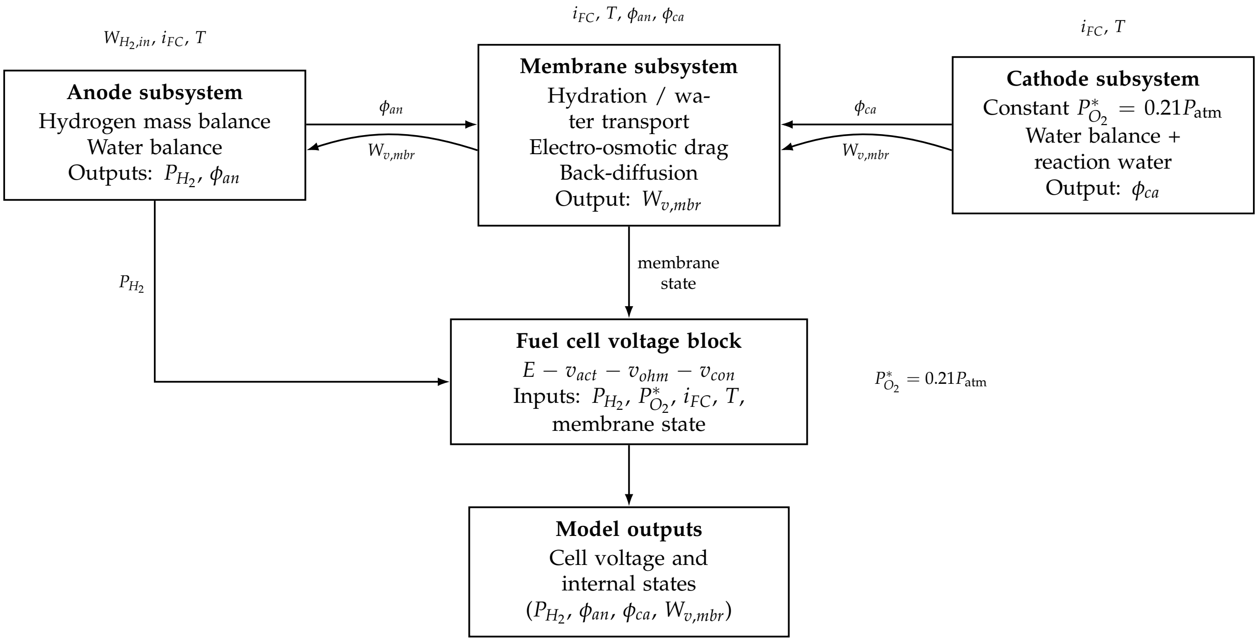


FIGURE 3.1: Block-level architecture of the simplified PEMFC model. The anode, membrane, and cathode subsystems exchange water and feed the voltage block, which combines the Nernst potential with the three losses to produce the cell voltage. The diagram makes explicit which signals are inputs (top) and which are internal states.

and the anode water balance keeps only the membrane flux, $\dot{m}_{w,an} = W_{v,mbr}$, with relative humidity $\phi_{an} = m_{w,an} R_v T / (V_{an} p_{sat}(T))$. The hydrogen pressure P_{H_2} enters the Nernst term of (3.1) and is the primary state of interest.

3.1.4 Membrane and Cathode

Water crosses the membrane mainly by electro-osmotic drag — protons dragging water molecules from anode to cathode — $W_{v,mbr} = M_v A_{fc}\, n\, n_d\, i_{FC}/F$ [9]; the membrane water content follows the Springer sorption isotherm $\lambda(a) = 0.043 + 17.81a - 39.85a^2 + 36.0a^3$, averaged over the two faces. On the cathode side, the oxygen partial pressure is held at $P^*_{O_2} = 0.21\, P_{\text{atm}}$, which removes the oxygen state while keeping the cathode water balance, $\dot{m}_{w,ca} = -W_{v,mbr} + W_{v,gen}$ with $W_{v,gen} = M_v n\, i_{FC}/(2F)$, whose humidity feeds back into the ohmic loss through λ_m. The block-level architecture of the resulting model is shown in Figure 3.1.

3.1.5 Model Properties and Next Steps

The preceding subsections define a compact model with a clear structure. Its state vector is $\mathbf{x} = (P_{H_2}, m_{w,an}, m_{w,ca})$: the anode hydrogen pressure and the water masses stored on each side of the membrane. Its inputs are the load current i_{FC}, set by the external load and treated as a disturbance, and the hydrogen inlet flow $W_{H_2,in}$, the manipulated input. Its single measured output is the cell voltage of (3.2). Each cell carries six calibration parameters, $\boldsymbol{\theta} = (\xi_1, \xi_2, \xi_3, \xi_4, B, c)$.

The model is deliberately simplified for control. The constant-P_{O_2} assumption and the lumped volumes keep the order low, at the cost of ignoring spatial gradients, oxygen transients, and liquid-water accumulation [14, 15]. This makes it well suited to fast simulation, linearization, and controller design, while remaining accurate enough to match each measured cell, as the results of Chapter 4 confirm.

These properties set the agenda for the rest of the methodology. First, the six parameters are calibrated per cell (Section 3.2). Next, the model is linearized and its controllability and observability are analyzed (Section 3.3), which determines what can be controlled and observed. The experimental protocols (Section 3.4) then provide the dynamic data used to validate the model and to fix the timescales for the control design of Chapter 5.

TABLE 3.1: Decision variables of the per-cell identification problem (3.7), with their role in the loss equations (3.3)–(3.5) and the search bounds. The four ξ coefficients shape the activation overvoltage; B and c scale the concentration and contact-resistance losses.

Variable	Physical role	Unit	Search bounds
ξ_1	Activation: constant term (3.3)	V	$[-8.0,\ 2.0]$
ξ_2	Activation: temperature term	$\mathrm{V\,K^{-1}}$	$[0,\ 3\times10^{-2}]$
ξ_3	Activation: oxygen-concentration term	$\mathrm{V\,K^{-1}}$	$[-5\times10^{-3},\ 5\times10^{-3}]$
ξ_4	Activation: current term (≤ 0)	$\mathrm{V\,K^{-1}}$	$[-1\times10^{-2},\ 0]$
B	Concentration-loss coefficient (3.5)	V	$[0,\ 2.0]$
c	Contact / lumped ohmic resistance (3.4)	Ω	$[0,\ 1\times10^{-2}]$

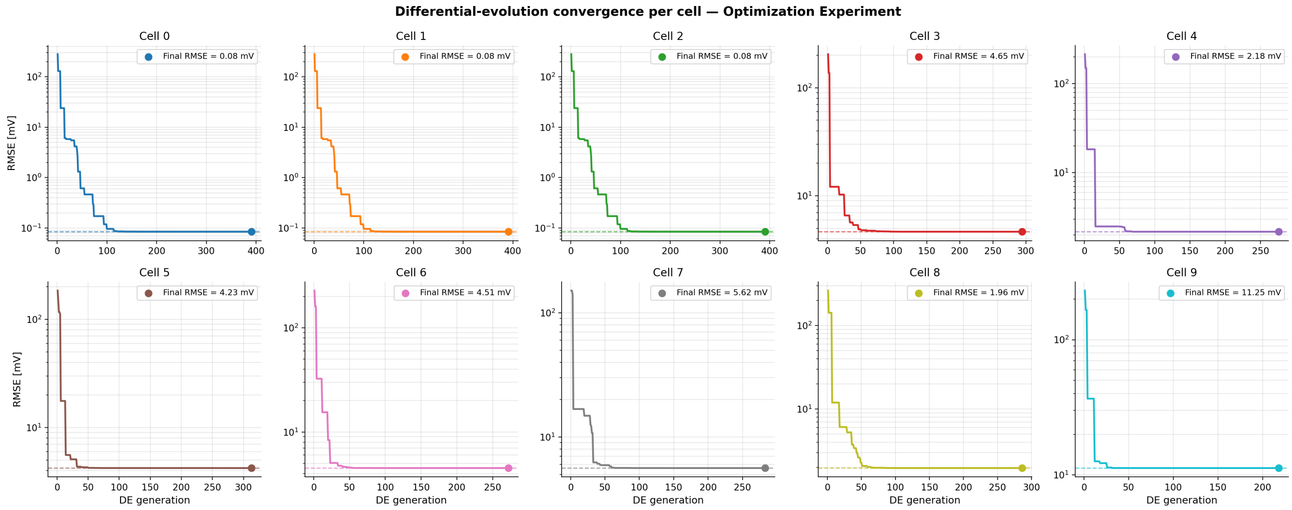


FIGURE 3.2: Differential-evolution convergence (RMSE versus generation, log scale) for the ten cells. The dashed line marks the final RMSE after L-BFGS-B refinement. Most cells converge within a few tens of generations; the degraded cells (0–2) collapse almost immediately to a near-zero RMSE against a flat zero-voltage signal, which is the first sign of a degenerate fit.

3.2 Per-Cell Parameter Identification

The six parameters of each cell are identified by minimizing the root-mean-square error (RMSE) between predicted and measured cell voltage over a calibration window:

$$\min_{\boldsymbol{\theta}} \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(v_{\text{model}}(I_i, P_{H_2,i};\boldsymbol{\theta}) - v_{\text{meas},i}\right)^2}. \tag{3.7}$$

Table 3.1 lists the decision variables, their physical role, and the box bounds used by the optimizer. The bounds also encode the physical constraints $\xi_4 \leq 0$ (activation overvoltage grows with current), $B \geq 0$, and $c \geq 0$ (losses are non-negative).

A two-stage strategy is used: **differential evolution** for the global search (population 20, up to 800 generations, fixed seed), followed by **L-BFGS-B** local refinement under the same bounds. Differential evolution is a population-based global optimizer that needs no gradient, which suits the non-convex objective; L-BFGS-B then polishes the best candidate. The optimizer converges for most cells within a few tens of generations (Figure 3.2).

3.3 State-Space Representation, Controllability and Observability

To prepare for control and observer design, the model is linearized around an operating point $(\mathbf{x}_0, \mathbf{u}_0)$ and written in the standard form $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$, $y = C\mathbf{x} + D\mathbf{u}$. Controllability is assessed through the rank of $\mathcal{C} = [B\ AB\ A^2B]$ and observability through $\mathcal{O} = [C;\ CA;\ CA^2]$.

3.3.1 State-Space Representation

The linearized model uses the state, input, and output vectors

$$\mathbf{x} = \begin{bmatrix} P_{H_2} \\ m_{w,an} \\ m_{w,ca} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} i_{FC} \\ W_{H_2,in} \end{bmatrix}, \qquad y = v_{fc},$$

where P_{H_2} is the anode hydrogen partial pressure, $m_{w,an}$ and $m_{w,ca}$ the water masses stored on the anode and cathode sides, i_{FC} the load current (a disturbance imposed by the load), $W_{H_2,in}$ the hydrogen inlet flow (the manipulated input), and v_{fc} the measured cell voltage. The four matrices have a direct physical reading: A describes the internal dynamics (water exchange across the membrane; the hydrogen pressure acts as an integrator), B describes how the current drawn and the inlet flow drive the states, C describes how the states appear in the measured voltage, and D is the direct feedthrough from the inputs to the voltage.

Matrices A and B — read directly from the dynamic equations

The three state equations of Section 3.1 are linear in the states and inputs once the temperature is held constant: the hydrogen pressure balance (3.6) (with $W_{H_2,out} \approx 0$ for the dead-end anode) and the two water balances written on the membrane water-transport terms,

$$\dot{m}_{w,an} = -\beta\, m_{w,an} + \gamma\, m_{w,ca} + d_1\, i_{FC}, \qquad \dot{m}_{w,ca} = \beta\, m_{w,an} - \gamma\, m_{w,ca} + d_2\, i_{FC},$$

with $\beta = D_w/(V_{an} t_m)$, $\gamma = D_w/(V_{ca} t_m)$, $d_1 = n_d M_v/F$, and $d_2 = (n - 2n_d)M_v/(2F)$. No linearization is needed; the matrices follow immediately:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\beta & \gamma \\ 0 & \beta & -\gamma \end{bmatrix}, \qquad B = \begin{bmatrix} -\dfrac{R_{H_2} T\, M_{H_2}\, n}{2F\, V_{an}} & \dfrac{R_{H_2} T}{V_{an}} \\ d_1 & 0 \\ d_2 & 0 \end{bmatrix}. \tag{3.8}$$

The first column of B is the effect of the load current (hydrogen consumption and electro-osmotic water displacement); the second column shows that the inlet flow acts on the hydrogen pressure only.

Matrices C and D — obtained by Jacobian linearization

Unlike the state equations, the output equation $y = v_{fc} = E(P_{H_2}) - v_{act}(i_{FC}) - v_{ohm}(i_{FC}, \lambda_m(m_{w,an}, m_{w,ca})) - v_{con}(i_{FC})$ is *nonlinear* in both states and inputs — it contains logarithms of P_{H_2} and i_{FC} and the nonlinear membrane-resistivity law (3.4) — so C and D cannot be read off the equations. They are obtained by a first-order Taylor (Jacobian) expansion around the operating point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$\delta y \;\approx\; \underbrace{\left.\frac{\partial v_{fc}}{\partial \mathbf{x}}\right|_{(\mathbf{x}_0,\mathbf{u}_0)}}_{C} \delta\mathbf{x} \;+\; \underbrace{\left.\frac{\partial v_{fc}}{\partial \mathbf{u}}\right|_{(\mathbf{x}_0,\mathbf{u}_0)}}_{D} \delta\mathbf{u}.$$

The three steps are:

1) *Hydrogen-pressure entry.* Only the Nernst term (3.1) depends on P_{H_2}: $\partial v_{fc}/\partial P_{H_2} = 4.31 \times 10^{-5}\, T/P_{H_2,0}$.

2) *Water-state entries.* The water masses act on the voltage through the chain $m_w \to \phi \to \lambda_m \to \sigma_m \to v_{ohm}$. Differentiating this chain gives

$$\frac{\partial v_{fc}}{\partial m_{w,an}} = \kappa_0\, \lambda'(\phi_{an,0})\, \frac{R_v T}{2V_{an}p_{sat}(T)}, \qquad \frac{\partial v_{fc}}{\partial m_{w,ca}} = \kappa_0\, \lambda'(\phi_{ca,0})\, \frac{R_v T}{2V_{ca}p_{sat}(T)},$$

with $\kappa_0 = i_0 t_m b_{11} \exp\left[b_2(\frac{1}{303} - \frac{1}{T})\right]/(A_{fc}\,\sigma_{m,0}^2)$ and $\lambda'(a) = 17.81 - 79.70a + 108a^2$ (derivative of the Springer isotherm).

3) *Input entries.* Differentiating the three loss terms with respect to i_{FC} at i_0, and noting that $W_{H_2,in}$ does not enter the output equation directly:

$$\frac{\partial v_{fc}}{\partial i_{FC}} = -\Big(\frac{\xi_4 T}{i_0} + \rho_{m,0}\frac{t_m}{A_{fc}} + c + \frac{B}{i_{\max} - i_0}\Big), \qquad \frac{\partial v_{fc}}{\partial W_{H_2,in}} = 0.$$

The resulting output matrices at the operating point are

$$C = \left[\frac{4.31 \times 10^{-5}\, T}{P_{H_2,0}} \quad \kappa_0\, \lambda'(\phi_{an,0})\, \frac{R_v T}{2V_{an}p_{sat}(T)} \quad \kappa_0\, \lambda'(\phi_{ca,0})\, \frac{R_v T}{2V_{ca}p_{sat}(T)}\right], \tag{3.9}$$

$$D = \left[-\Big(\frac{\xi_4 T}{i_0} + \rho_{m,0}\frac{t_m}{A_{fc}} + c + \frac{B}{i_{\max} - i_0}\Big) \quad 0\right]. \tag{3.10}$$

In the simplified model where λ_m is held constant, the two water-state entries of C vanish and $C_{\text{red}} = \left[4.31 \times 10^{-5}\, T/P_{H_2,0} \;\; 0 \;\; 0\right]$.

3.3.2 Controllability

Because P_{H_2} is directly driven by $W_{H_2,in}$ (entry $R_{H_2}T/V_{an} > 0$ in B), controllability of the full system reduces to controllability of the two-state water subsystem (A_w, B_w) with $A_w = \left[\begin{smallmatrix} -\beta & \gamma \\ \beta & -\gamma \end{smallmatrix}\right]$ and $B_w = [d_1; d_2]$ from (3.8). Its controllability determinant, $\det(\mathcal{C}_w) = (d_1 + d_2)(\beta d_1 - \gamma d_2)$ with $d_1 + d_2 = nM_v/(2F) > 0$, vanishes only under the single degenerate condition

$$n_d = \frac{n\, V_{an}}{2\,(V_{an} + V_{ca})}, \tag{3.11}$$

i.e. when electro-osmotic drag exactly cancels the back-diffusion transport. **The system is therefore fully controllable for all physically realistic drag coefficients.**

3.3.3 Observability

With voltage as the only measurement, the observability matrix $\mathcal{O} = [C; CA; CA^2]$ is built from (3.9). In the simplified model the output reduces to C_{red}, so the **voltage observes the hydrogen pressure but not the two water states.** In the full model all three states are potentially observable, but the water-state sensitivities in (3.9) are small and degrade away from the operating point. The practical consequence is that hydrogen-pressure regulation can rely on a reduced-order voltage-based observer, whereas a full-state observer would require additional pressure or humidity sensors.

3.4 Experimental Protocols

All experiments were performed on the ten-cell educational PEMFC stack of the laboratory bench, with per-cell voltage acquisition at 1 Hz through the serial data-acquisition chain developed in the project.

- **Optimization Experiment** (April 2026): a fixed 25 Ω resistor is connected at $t = 135$ s and removed at $t = 684$ s, with no valve-driven current control. The calibration points used for the per-cell parameter identification are drawn from the loaded window of this experiment.

- **TR-current step experiments** (9 June 2026): the stack is left at open circuit, then a fixed resistor ($3.3\,\Omega$ for Experiment 1, $6.8\,\Omega$ for Experiment 4) is connected at $t = 0$. The processing pipeline detects the connection event from a $> 100\,\text{mA}$ current jump and re-zeros the time axis. The current step response is fitted with the exponential model $I(t) = I_{\text{ss}} + \Delta I\, e^{-t/\tau_I}$, from which settling times $T_{s,p\%} = -\tau_I \ln(p/100)$ are derived analytically; the $90\,\% \rightarrow 10\,\%$ fall time is measured on the raw signal.

Chapter 4

Results & Analysis

4.1 Parameter Identification on the Optimization Experiment

The two-stage optimizer of Section 3.2 identified the six parameters for all ten cells from the loaded window of the Optimization Experiment. The complete set of tuned values is given in Appendix B, Table B.1. For the healthy cells the calibration RMSE ranges from 2.0 to 11.3 mV — below 12 mV in every case — and the mean time-series RMSE over the healthy cells is 8.0 mV. These figures confirm that a six-parameter law reproduces each cell to within a few millivolts.

Figure 4.1 places the experiment in context. The stack sits at its open-circuit plateau of about 4.57 V, drops to roughly 0.8 V when the 25 Ω resistor is connected at $t = 135$ s, draws a load current that decays from about 30 mA, and recovers when the resistor is removed at $t = 684$ s. The simulated stack voltage (sum of the ten calibrated cells) tracks the measured trace throughout the loaded window, which shows that the per-cell fits aggregate correctly to the stack level.

The per-cell calibration fits in Figure 4.2 explain these numbers. Two findings stand out. First, the model describes the healthy cells accurately with only six parameters: the activation coefficients are consistent across cells, while the concentration (B) and contact-resistance (c) terms stay close to zero, indicating that activation and ohmic effects dominate over the covered range. Second, the identification doubles as a *diagnosis* tool. Cells 0, 1, and 2 collapse to about 0 V under load, and the optimizer returns identical, degenerate parameter sets for them (RMSE below 0.1 mV against a flat near-zero signal). This is the signature of a non-functioning electrode, not a physically meaningful fit, because no parameter combination within the bounds can reproduce a dead cell.

The time-series comparison in Figure 4.3 confirms the fit cell by cell. The simulated voltage overlays the measurement for the healthy cells across the loaded window, whereas Cells 0–2 remain pinned near zero. The slightly larger residuals of Cells 7 and 9 stem from the load-on transient, which the calibration window deliberately excludes; over the settled window the agreement is within the reported RMSE.

4.2 Step-Response Characterization (TR-Current Experiments)

The TR-current experiments characterize how fast the stack current settles after a load step. In both runs, Cells 1 and 2 read about 0 mV throughout the loaded window, confirming the degradation first seen in the Optimization Experiment; the other cells behave normally. Figure 4.4 shows the raw channels for the two loads.

Fitting the first-order model $I(t) = I_{ss} + \Delta I\, e^{-t/\tau_I}$ to the current gives the metrics of Table 4.1, illustrated in Figure 4.5.

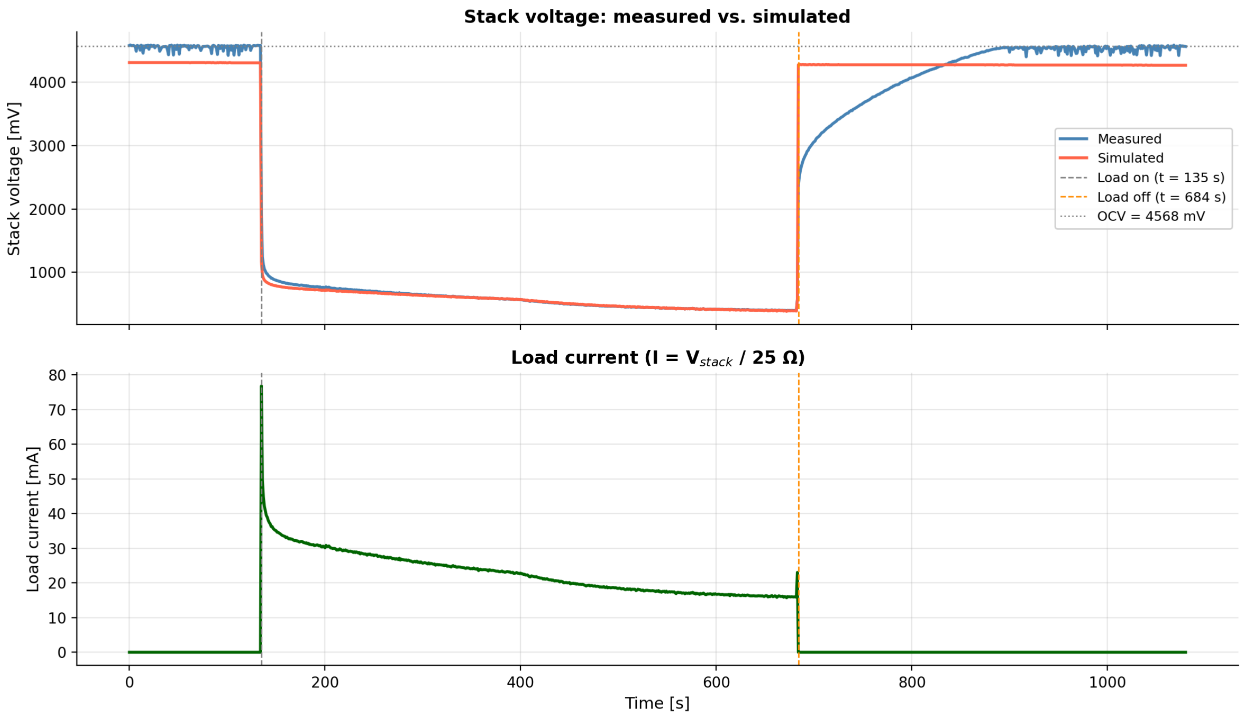


FIGURE 4.1: Stack voltage and load current during the Optimization Experiment. **Top:** measured (blue) and simulated (red) stack voltage; the load is applied at $t = 135\,\text{s}$ and removed at $t = 684\,\text{s}$. **Bottom:** the load current $I = V_{\text{stack}}/25\,\Omega$. The close overlap in the loaded window confirms that the calibrated cell models sum to the correct stack voltage.

TABLE 4.1: Current step-response metrics for the two loads. The lighter load ($6.8\,\Omega$) gives a longer time constant and a smaller initial perturbation, consistent with slower dynamics at lower current density.

Current metric	**Exp 1 (3.3 Ω)**	**Exp 4 (6.8 Ω)**
Steady-state current I_{ss} [mA]	572.9	556.1
Step amplitude ΔI [mA]	82.1	52.9
Time constant τ_I [s]	8.2	10.3
Settling time $T_{s,2\%}$ [s]	32.0	40.2
Fall time 90%→10% [s]	16.0	19.0

The current time constant is about 25 % longer under the lighter load ($\tau_I = 10.3\,\text{s}$ at $6.8\,\Omega$ versus $8.2\,\text{s}$ at $3.3\,\Omega$), consistent with slower electrochemical dynamics at lower current density; the initial perturbation ΔI is also smaller (52.9 versus 82.1 mA). The settling times derived from the fit ($T_{s,2\%} = 32$–$40\,\text{s}$) are robust to measurement noise, because they are computed analytically from τ_I rather than read off the noisy raw signal. This timescale captures the dominant dynamics seen by the load — electrochemical kinetics combined with gas-supply transients — and sets a quantitative target for controller bandwidth: the closed-loop controller of Chapter 5 must act well within $\tau_I \approx 8$–$10\,\text{s}$ to shape the stack's transient behavior.

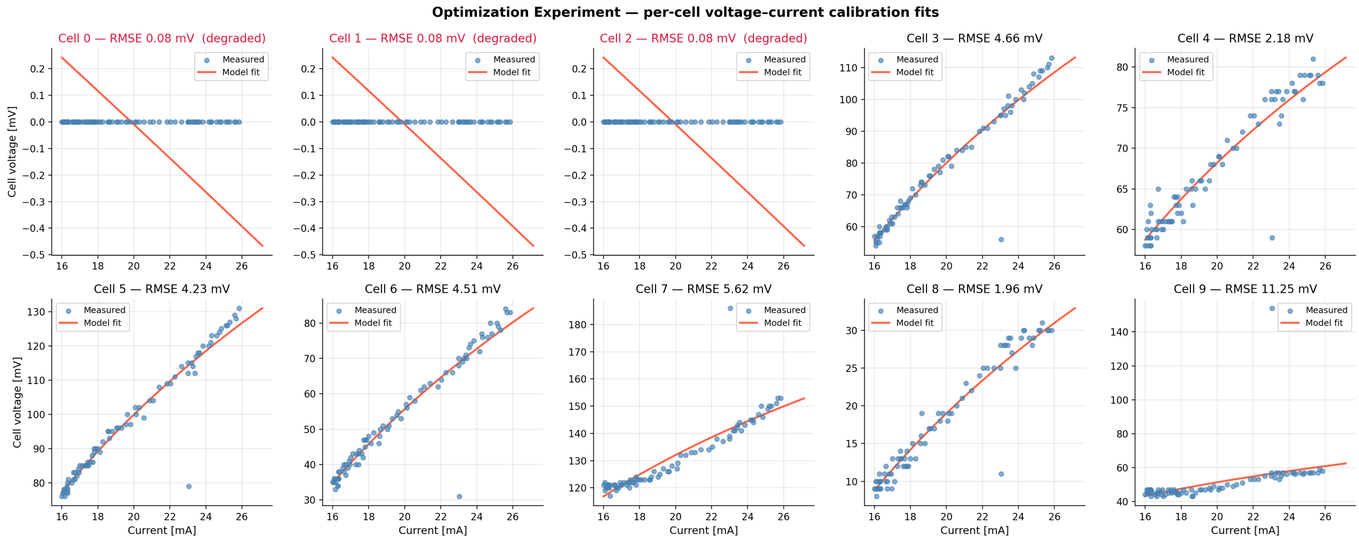


FIGURE 4.2: Per-cell voltage–current calibration fits: measured points (blue) versus the identified model (red). The seven healthy cells (3–9) are matched with RMSE between 2.0 and 11.3 mV. Cells 0–2 (highlighted) collapse to a near-zero voltage under load, so the optimizer cannot produce a meaningful curve — a clear signature of hardware degradation.

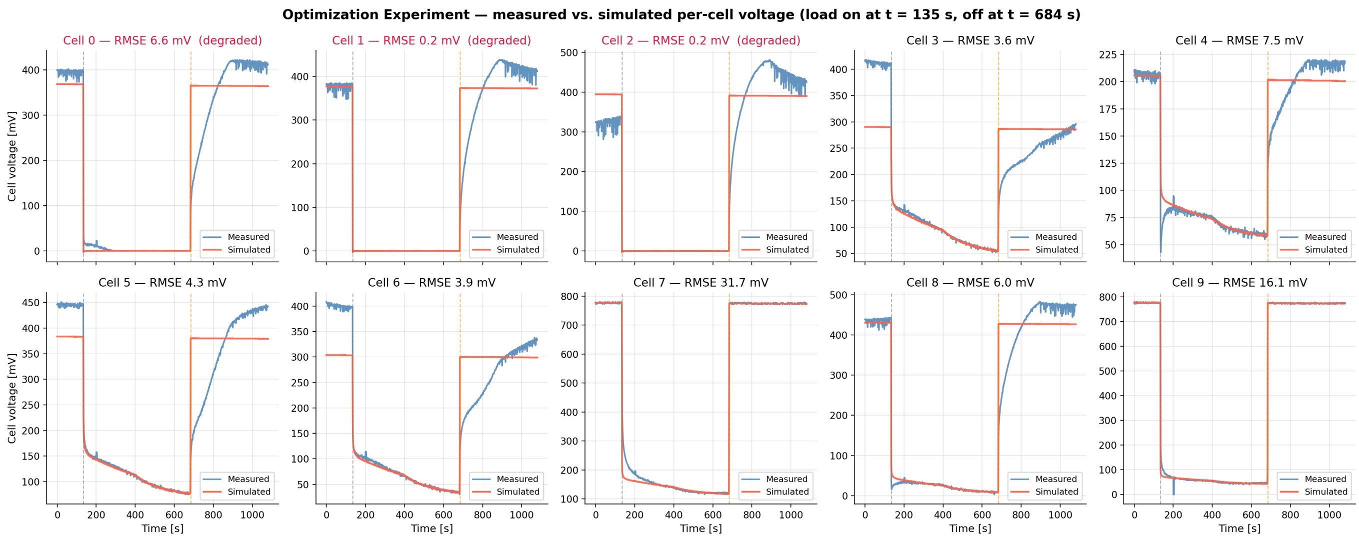
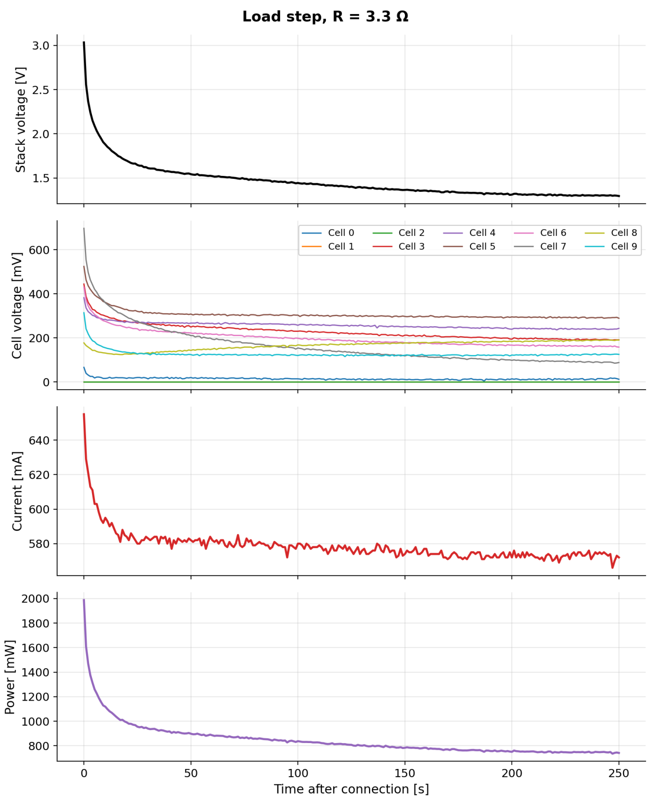
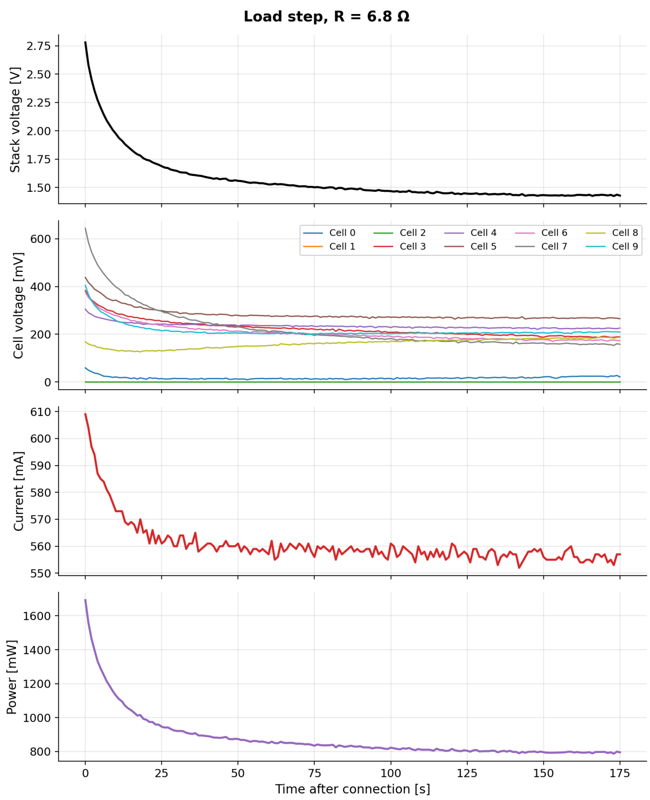


FIGURE 4.3: Per-cell voltage, measured (blue) versus simulated (red), over the full experiment. Healthy cells (3–9) are tracked closely; the degraded cells 0–2 stay near zero under load. Titles report the time-series RMSE over the loaded window (mean 8.0 mV for the healthy cells).



(A) Experiment 1, $R = 3.3\,\Omega$.



(B) Experiment 4, $R = 6.8\,\Omega$.

FIGURE 4.4: TR-current step experiments after resistor connection (time re-zeroed to the step): stack voltage, individual cell voltages, current, and power. Both signals relax monotonically toward a steady state, with no overshoot — behavior well captured by a first-order model.

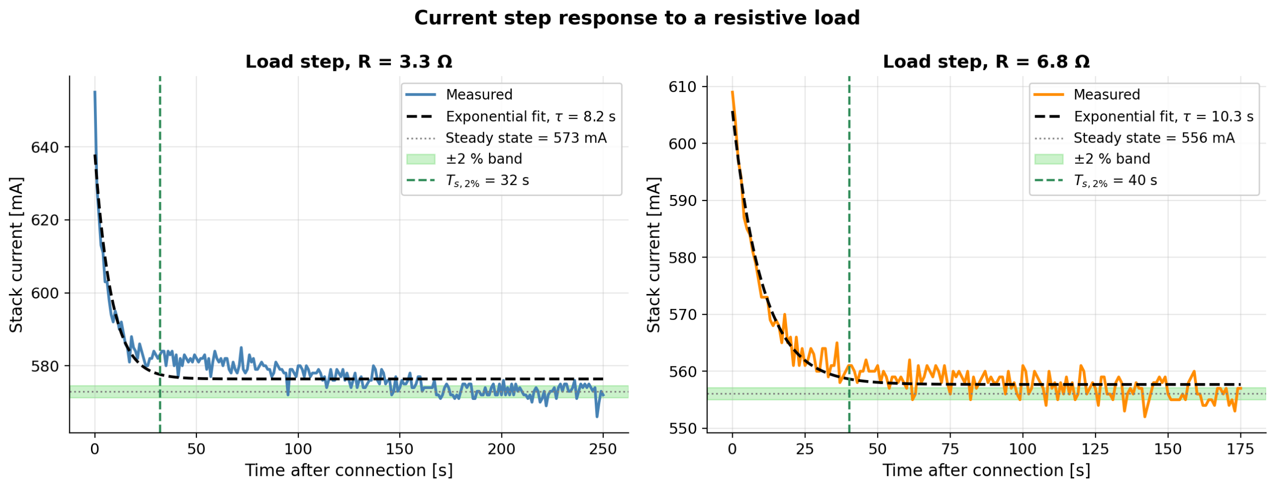


FIGURE 4.5: Current step response for the two loads: measured current (color), exponential fit (dashed), steady-state level, $\pm 2\,\%$ settling band, and the $2\,\%$ settling time. The exponential model matches the data closely, justifying the first-order time constants reported in Table 4.1.

Chapter 5

Control of the PEMFC Power System: DC/DC Boost Converter with PI Controller

This chapter presents the control architecture adopted for the PEMFC power system. As motivated in Section 2.3.1, the hydrogen supply offers too little authority to regulate the load directly, so control is applied on the electrical side through a DC/DC converter. The design follows the approach of Derbeli et al. [4].

5.1 System Overview

By itself, a PEMFC stack is *neither* an ideal voltage source *nor* an ideal current source: its terminal voltage V_{STACK} is low and varies strongly with the current drawn, following the polarization curve of Chapter 2. The role of the control stage is to shape this raw output into a *regulated source*. The stack feeds a **DC/DC boost converter** (a step-up chopper), and a **PI controller** closes the loop by acting on the converter duty cycle T through pulse-width modulation (PWM) [4]. Figure 5.1 shows the architecture.

Depending on which variable the PI loop regulates, the *converter output* — i.e. the PEMFC power system seen from the load — can be made to behave as:

- a **controlled current source** — the loop regulates the inductor (stack) current I_L to a reference I_{REF} (Section 5.2);
- a **controlled voltage source** — the loop regulates the output voltage V_{OUT} to a reference V_{REF} (Section 5.3).

It is therefore the regulated converter output, not the fuel cell itself, that is made to act as a current or a voltage source.

5.2 Output Current Regulation (Current-Source Operation)

Here the goal is to make the regulated converter output behave as a **current source**. Following [4], the design proceeds in five steps.

1) *Control objective.* The controlled variable is the inductor (stack) current I_L, which the PI loop forces to track a reference I_{REF} so that the system delivers a regulated current. In the reference design, $I_{REF} = 9.74\,\text{A}$ is the maximum-power operating point of the 50 W, ten-cell stack used for validation (54 W at 5.54 V). Regulating the current directly also protects the stack, whose maximum current (10 A) must never be exceeded.

2) *Boost converter state-space model.* Averaging the two switching topologies of the converter (switch ON / switch OFF) over one PWM period yields, with state $x = [\,i_L \quad V_{OUT}\,]^\top$, duty cycle T, load R, inductance L, and capacitance C [4]:

$$\dot{x} = \begin{bmatrix} 0 & \dfrac{T-1}{L} \\ \dfrac{1-T}{C} & -\dfrac{1}{RC} \end{bmatrix} x + \begin{bmatrix} \dfrac{1}{L} \\ 0 \end{bmatrix} u, \qquad y = \begin{bmatrix} 0 & 1 \end{bmatrix} x. \tag{5.1}$$

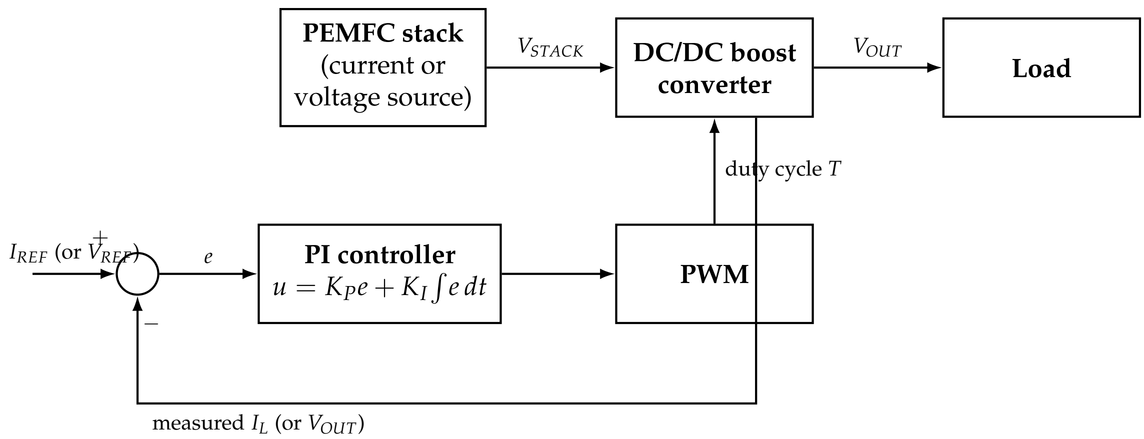


FIGURE 5.1: Control architecture of the PEMFC power system, after [4]: the stack feeds a DC/DC boost converter whose duty cycle is set by a PI controller. Regulating the inductor current I_L makes the converter output behave as a current source; regulating the output voltage V_{OUT} makes it behave as a voltage source.

3) *PI control law.* The error signal is $e = I_{REF} - I_L$, and the control output (duty cycle) combines proportional and integral action:

$$u = K_P\, e + K_I \int e\, dt. \tag{5.2}$$

The proportional term shapes the rise time and the integral term removes the steady-state error, at the cost of possible overshoot if the gains are too aggressive.

4) *PI tuning (online method).* The gains are tuned experimentally, without knowledge of the plant transfer function [4]: set $K_P = 0$ and $T_i = \infty$; increase K_P until sustained oscillations appear and note this value K_{PU}; set $K_P = K_{PU}/2$; then decrease T_i until oscillations reappear (T_{iU}) and set $T_i = 2\,T_{iU}$.

5) *Expected behavior.* With properly tuned gains, the PI controller produces a smooth, stable rise of I_L to I_{REF} ($\approx$ 400 ms in the reference design) and good disturbance rejection under large load variations, as validated on a dSPACE 1104 real-time platform [4].

5.3 Output Voltage Regulation (Voltage-Source Operation)

Here the goal is to make the same regulated converter output behave instead as a **voltage source**, using the same PI-based methodology.

1) *Control objective.* The controlled variable is now the converter output voltage V_{OUT}, which the PI loop forces to track a reference V_{REF} so that the system delivers a regulated voltage to the load.

2) *Boost converter model.* The same averaged state-space model (5.1) applies; the output of interest is the second state, $y = V_{OUT}$.

3) *PI control law.* The error signal becomes $e = V_{REF} - V_{OUT}$, and the same control law (5.2) applies.

4) *Tuning.* The same online procedure is used, adapted to the voltage-loop dynamics, which include the RC output stage and are therefore slower than the current loop.

5) *Comparison of the two objectives.* Regulating the current is better suited to protecting the PEMFC from overcurrent, since the controlled variable is the current drawn from the stack; regulating the output voltage is more natural for supplying a fixed voltage to the load. Both make the converter output behave as a regulated source, rely on the same PI structure and converter model, and differ only in the feedback variable (I_L vs. V_{OUT}) and in the physical meaning of the reference.

Chapter 6

Conclusion & Perspectives

This work developed a simplified control-oriented model of a ten-cell PEMFC stack, calibrated it per cell on the Optimization Experiment, derived and analyzed its linearized state-space representation, characterized its dynamic current response to load steps, and proposed a PI-based control architecture for the PEMFC power system.

The main outcomes are: (i) a six-parameter semi-empirical voltage model coupled to lumped anode, membrane, and cathode dynamics that fits all healthy cells to within 12 mV; (ii) a diagnosis capability — degraded cells (0, 1, 2) are revealed both by voltage collapse and by degenerate, physically meaningless fits; (iii) an explicit state-space representation in which A and B follow directly from the dynamic equations while C and D require Jacobian linearization of the nonlinear voltage equation, leading to proof that the model is generically controllable while a voltage-only observer can reconstruct the hydrogen pressure but not the water states; (iv) an experimental characterization of the current response to load steps ($\tau_I \approx$ 8–10 s, $T_{s,2\%} \approx$ 32–40 s); and (v) a control architecture based on a DC/DC boost converter regulated by a PI controller [4], motivated by the limited authority of the hydrogen supply, in which the converter output is regulated to behave either as a current source (overcurrent protection) or as a voltage source (output-voltage regulation).

Perspectives follow directly. On the control side, the next step is to implement the proposed boost-converter PI loop on the bench and tune it online following [4], with the identified current timescale bounding the required closed-loop bandwidth; the observability analysis further supports a hydrogen-pressure regulator with a reduced-order voltage-based observer, with PI control as the literature-standard baseline [16, 4] and LQ or time-delay control as upgrades [9, 7]. On the modeling side, capturing oxygen transients (removing the constant-P_{O_2} assumption), liquid-water accumulation, and gas-supply delays [6] would extend validity at higher dynamics. Finally, replacing the degraded cells and repeating the calibration would allow stack-level closed-loop experiments.

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Appendix A

Derivation of the Open-Circuit Voltage

The global hydrogen–oxygen reaction is $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$. At non-standard conditions the Gibbs free energy change is

$$\Delta g_f = \Delta g_f^0 - \bar{R}T_{fc} \ln\left(\frac{p_{H_2}\, p_{O_2}^{1/2}}{p_{H_2O}}\right),$$

and for a reversible cell $\Delta g_f = -2FE$, so

$$E = -\frac{\Delta g_f^0}{2F} + \frac{\bar{R}T_{fc}}{2F} \ln\left(\frac{p_{H_2}\, p_{O_2}^{1/2}}{p_{H_2O}}\right).$$

Standard Voltage

Taking liquid water as the product at 25 °C:

$$E^0 = -\frac{\Delta g_f^0}{2F} = -\frac{-237{,}200}{2 \times 96{,}485} \approx 1.229 \text{ V}.$$

Table A.1 lists Δg_f^0 at several temperatures.

TABLE A.1: Gibbs free energy change of the hydrogen fuel cell reaction at various temperatures [9].

Product	**Temperature (°C)**	**Δg_f^0 (kJ/mol)**
Liquid	25	−237.2
Liquid	80	−228.2
Gas	80	−226.1
Gas	100	−225.2
Gas	200	−220.4
Gas	600	−199.6
Gas	1000	−177.4

Temperature Coefficient

From $\Delta G^0 = \Delta H^0 - T\Delta S^0$, with

$$\Delta S^0 = S^0_{H_2O(l)} - S^0_{H_2(g)} - \tfrac{1}{2}S^0_{O_2(g)} = 69.95 - 130.7 - 102.58 \approx -163.3 \text{ J mol}^{-1}\text{K}^{-1}$$

for the liquid-water reaction,

$$\frac{\Delta S^0}{2F} \approx \frac{-163.3}{192{,}970} \approx -0.85 \times 10^{-3} \text{ V K}^{-1}.$$

Nernst Pressure Correction

Assuming $p_{H_2O} = 1$ (liquid product):

$$\frac{\bar{R}T_{fc}}{2F}\ln\bigl(p_{H_2}\,p_{O_2}^{1/2}\bigr) = \frac{8.314\,T_{fc}}{192{,}970}\left[\ln p_{H_2} + \tfrac{1}{2}\ln p_{O_2}\right] \approx 4.31\times10^{-5}\,T_{fc}\left[\ln p_{H_2} + \tfrac{1}{2}\ln p_{O_2}\right].$$

Final Expression

Combining all terms yields Equation (3.1) of the main text:

$$E = 1.229 - 0.85\times10^{-3}(T_{fc} - 298.15) + 4.31\times10^{-5}\,T_{fc}\bigl[\ln p_{H_2} + \tfrac{1}{2}\ln p_{O_2}\bigr]\ \text{V}.$$

Appendix B

Detailed Tables

TABLE B.1: Tuned parameters and calibration RMSE for all ten cells — Optimization Experiment. Cells 0–2 collapsed under load: the optimizer returns identical degenerate parameter sets that reproduce a near-zero voltage and are not physically meaningful.

Cell	RMSE [mV]	ξ_1	ξ_2	ξ_3	ξ_4	B	c
Cell 0[†]	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}	1.00×10^{-2}
Cell 1[†]	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}	1.00×10^{-2}
Cell 2[†]	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}	1.00×10^{-2}
Cell 3	4.66	-1.977	2.83×10^{-2}	1.32×10^{-3}	-3.17×10^{-4}	≈ 0	≈ 0
Cell 4	2.18	0.386	1.43×10^{-2}	8.11×10^{-4}	-1.24×10^{-4}	≈ 0	≈ 0
Cell 5	4.23	0.682	3.00×10^{-2}	1.93×10^{-3}	-2.98×10^{-4}	≈ 0	≈ 0
Cell 6	4.51	1.775	4.62×10^{-3}	4.87×10^{-4}	-2.73×10^{-4}	≈ 0	≈ 0
Cell 7	5.62	1.796	2.54×10^{-2}	1.82×10^{-3}	-1.99×10^{-4}	≈ 0	≈ 0
Cell 8	1.96	-4.946	2.81×10^{-2}	6.92×10^{-4}	-1.34×10^{-4}	≈ 0	≈ 0
Cell 9	11.25	-5.400	2.01×10^{-2}	9.31×10^{-5}	-1.06×10^{-4}	3.21×10^{-7}	≈ 0

[†] Degraded cell (voltage collapse under load); fit not physically meaningful.